WORKED SOLUTION

STEP II 2019, Question 1

Pure • Tangents to Cubics

STEP II 2019, Question 1

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

Write down the condition for the tangent at $x = a$ to pass through $(p, 0)$ and substitute $f(x) = (x - p)g(x)$: almost everything cancels, leaving a multiple of $g'(a)$. For the cubic, the tangent from the root p therefore touches the curve where the *other* quadratic factor is stationary, which is midway between the other two roots. Part (ii) then comes down to comparing two gradients, and to factorising the equation for where a tangent meets a cubic, which always has a double root at the point of contact.

Solution

The opening result

The tangent at $x = a$ is $y = f(a) + f'(a)(x - a)$. By the product rule, $f'(x) = g(x) + (x - p)g'(x)$. The tangent passes through $(p, 0)$ exactly when

$$\begin{aligned} 0 &= f(a) + f'(a)(p - a) \\ &= (a - p)g(a) + [g(a) + (a - p)g'(a)](p - a) \\ &= (a - p)g(a) - (a - p)g(a) - (a - p)^2 g'(a) \\ &= -(a - p)^2 g'(a). \end{aligned}$$

Since $a \neq p$, the factor $(a - p)^2$ is non-zero, so this holds if and only if $g'(a) = 0$. That proves the result in both directions at once.

Part (i)

Here $f(x) = (x - p)g(x)$ with $g(x) = A(x - q)(x - r) = A(x^2 - (q + r)x + qr)$, so $g'(x) = A(2x - (q + r))$. By the opening result, and since $A \neq 0$,

$$g'(a) = 0 \iff 2a = q + r.$$

(Note that $a = \frac{1}{2}(q + r) > q > p$, so $a \neq p$ as required.)

For the gradient, $f'(a) = g(a) + (a-p)g'(a) = g(a)$ because $g'(a) = 0$. With $a - q = \frac{1}{2}(r-q)$ and $a - r = -\frac{1}{2}(r-q)$,

$$f'(a) = A \cdot \tfrac{1}{2}(r-q) \cdot (-\tfrac{1}{2}(r-q)) = -\frac{A(r-q)^2}{4}.$$

Part (ii)

Now factor out the root r instead: $f(x) = (x-r)h(x)$ with $h(x) = A(x-p)(x-q)$. The opening result applies with r in place of p (nothing in it used which root we chose), so the tangent at $x = c$ passes through $(r, 0)$ if and only if $h'(c) = 0$, that is $c = \frac{1}{2}(p+q)$. Exactly as in part (i), its gradient is

$$f'(c) = h(c) = A \cdot \tfrac{1}{2}(q-p) \cdot (-\tfrac{1}{2}(q-p)) = -\frac{A(q-p)^2}{4}.$$

So, since $A \neq 0$,

$$\text{tangents parallel} \iff (r-q)^2 = (q-p)^2 \iff r - q = q - p \iff q = \tfrac{1}{2}(p+r),$$

where the middle step uses $r - q > 0$ and $q - p > 0$, so we may take positive square roots.

Ask yourself How do we find where a tangent meets a cubic again, without solving a cubic?

We know one root already, twice over: the point of contact. At $x = q$ the gradient is $f'(q) = A(q-p)(q-r)$ (the other two terms of the product rule vanish because they contain $x - q$), and $f(q) = 0$, so the tangent there is $y = A(q-p)(q-r)(x-q)$. It meets the curve where

$$A(x-p)(x-q)(x-r) = A(q-p)(q-r)(x-q).$$

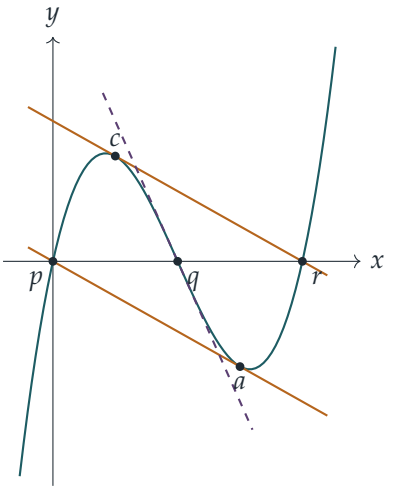
Dividing by $A \neq 0$ and taking out the factor $x - q$,

$$\begin{aligned} 0 &= (x-q)[(x-p)(x-r) - (q-p)(q-r)] \\ &= (x-q)[x^2 - q^2 - (p+r)(x-q)] \\ &= (x-q)^2\,(x+q-p-r). \end{aligned}$$

So the tangent meets the curve at $x = q$ (a double root, as expected for a point of contact) and at $x = p + r - q$, and nowhere else. It fails to meet the curve again if and only if this third root coincides with q:

$$p + r - q = q \iff q = \tfrac{1}{2}(p+r).$$

This is the same condition as before, so the tangents of parts (i) and (ii) are parallel if and only if the tangent at $x = q$ does not meet the curve again.



The sketch shows the case $q = \frac{1}{2}(p + r)$: the two tangents (amber) are parallel, and the tangent at q (dashed) crosses the curve there without meeting it anywhere else, since q is then the point of inflection.

Checks

- Take $p = 0$, $q = 1$, $r = 2$, $A = 3$, so $f(x) = 3x^3 - 9x^2 + 6x$. Then $f'(x) = 9x^2 - 18x + 6$, giving $f'(\frac{3}{2}) = -\frac{3}{4}$ and $f'(\frac{1}{2}) = -\frac{3}{4}$, both equal to $-A(r-q)^2/4 = -A(q-p)^2/4 = -\frac{3}{4}$, so the tangents are parallel, as predicted since q is the midpoint of p and r.
- The tangent at $x = \frac{3}{2}$ is $y = -\frac{9}{8} - \frac{3}{4}(x - \frac{3}{2})$, which is 0 at $x = 0$: it does pass through $(p, 0)$.
- All the identities above were also confirmed by computer in exact arithmetic for about two thousand random choices of $p < q < r$ and A.

Common slips

- Forgetting that $a \neq p$ is what allows us to cancel $(a - p)^2$; without it the "only if" direction fails, since the tangent at p itself always passes through $(p, 0)$.
- Writing $(r - q)^2 = (q - p)^2 \Rightarrow r - q = q - p$ without saying why the negative root is excluded.
- Trying to solve the cubic for the intersections from scratch, instead of using the known double root at the point of contact.

Beyond STEP

The tangent at $x = q$ is special because q is then the inflection point, the centre of symmetry of the cubic: every cubic is symmetric under a half-turn about its inflection point, which is why the two tangents become parallel. The tangent and chord constructions here are the first steps towards the geometry of cubic curves, where the third intersection of a line with a cubic is used to define an addition law on its points (elliptic curves). Siklos, *Advanced Problems in Mathematics*, has more questions in this spirit.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.